# Distributions: A Working Reference

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For each distribution: density/mass, support, moments, the MLE, and its sufficient statistic. `distributions.py` verifies every moment formula empirically. More entries will be added as needed.

## 1 Bernoulli$(p)$

$x \in \{0,1\}$, parameter $p \in (0,1)$:

$$\Pr(x) = p^x(1-p)^{1-x}, \qquad \mathbb{E}[x] = p, \qquad \mathrm{Var}(x) = p(1-p). \tag{1}$$

Variance is maximal at $p = \frac{1}{2}$ and vanishes at the deterministic endpoints. MLE from $n$ i.i.d. samples: $\hat{p} = m/n$ with $m = \sum_i x_i$ (derived in `bernoulli/`); $m$ is sufficient. Exponential-family form with natural parameter the log-odds $t = \log \frac{p}{1-p}$ (the logit — the reason the sigmoid keeps appearing).

## 2 Multinoulli / categorical$(p_1, \ldots, p_k)$

One draw among $k$ outcomes, $\sum_j p_j = 1$. With the one-hot encoding $e_x \in \{0,1\}^k$:

$$\Pr(x = j) = p_j, \qquad \mathbb{E}[e_x] = p, \qquad \mathrm{Cov}(e_x)_{ij} = \begin{cases} p_i(1-p_i) & i = j \\ -\,p_i p_j & i \neq j. \end{cases} \tag{2}$$

The negative off-diagonal covariance is structural: outcomes compete — one category occurring excludes the others. Over $n$ draws the counts $(n_1, \ldots, n_k)$ are multinomial and sufficient; the MLE (a constrained problem solved with a Lagrange multiplier, cf. the optimization note) is

$$\hat{p}_j \;=\; \frac{n_j}{n}. \tag{3}$$

Softmax is to the multinoulli what the sigmoid is to the Bernoulli: the map from unconstrained scores to the probability simplex.

## 3 Normal $\mathcal{N}(\mu, \sigma^2)$

$x \in \mathbb{R}$:

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\!\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big), \qquad \mathbb{E}[x] = \mu, \quad \mathrm{Var}(x) = \sigma^2. \tag{4}$$

Facts to keep at hand: closed under affine maps ($ax+b \sim \mathcal{N}(a\mu+b,\, a^2\sigma^2)$); sums of independent normals are normal; 68/95/99.7% of mass within 1/2/3 standard deviations; maximizes entropy among densities on $\mathbb{R}$ with fixed variance (information-theory note); the CLT makes it the universal limit of standardized sums. MLE: $\hat{\mu} = \bar{x}$, $\hat{\sigma}^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$ — biased by design (`gaussian/`); $(\sum x_i, \sum x_i^2)$ is sufficient.

## 4 Multivariate normal $\mathcal{N}(\mu, \Sigma)$

$x \in \mathbb{R}^d$, mean $\mu \in \mathbb{R}^d$, covariance $\Sigma \succ 0$ symmetric:

$$f(x) = \frac{1}{(2\pi)^{d/2}\,|\Sigma|^{1/2}} \exp\Big(-\tfrac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\Big). \tag{5}$$

$\mathbb{E}[x] = \mu$, $\mathrm{Cov}(x) = \Sigma$. Level sets are ellipsoids whose axes are the eigenvectors of $\Sigma$ with half-lengths $\propto \sqrt{\lambda_i}$ (drawn by the script). Closed under affine maps: $Ax + b \sim \mathcal{N}(A\mu + b,\, A\Sigma A^\top)$; all marginals and conditionals are again normal — the property that makes Gaussians the workhorse of tractable multivariate modeling. For Gaussians (and only them, among joint normals) uncorrelated $\Rightarrow$ independent. MLE from $n$ samples:

$$\hat{\mu} = \bar{x}, \qquad \hat{\Sigma} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(x_i - \bar{x})^\top, \tag{6}$$

again the $1/n$ (biased) form.

## To be added

Poisson, exponential, uniform, Laplace, Beta/Dirichlet (conjugate priors for the two Bernoulli-family entries above), Student-$t$.